Glycolysis and Fermentation Presentation Script

Unit 3B Slides 8-16

Slide 8 Step 1 Glycolysis

We are now focusing on the first stage of cellular respiration: glycolysis. The name tells us what is happening. Glyco refers to sugar, and lysis means to split. During glycolysis, one six-carbon glucose molecule is rearranged and divided into two three-carbon molecules called pyruvate.

Glycolysis is the starting point for both aerobic and anaerobic pathways. It does not require oxygen directly, and it occurs in the cytosol rather than inside the mitochondrion. That means nearly every organism can perform glycolysis, including prokaryotes, which lack mitochondria.

Slide 9 Glycolysis Overview

Glycolysis is a ten-step, enzyme-controlled pathway in the cytosol. The overall result is that one glucose becomes two pyruvate molecules. Because glucose has six carbons and each pyruvate has three, all six carbon atoms are still present at the end of glycolysis. No carbon dioxide is released during this stage.

The pathway is anaerobic because oxygen is not one of its reactants. Glycolysis can therefore occur whether oxygen is present or absent. However, it captures only a small portion of glucose's available energy. The net yield is two ATP per glucose, which is why the slide describes it as inefficient compared with complete aerobic respiration. A considerable amount of energy remains stored in the two pyruvate molecules and in NADH.

Do not confuse the number of ATP produced with the net gain. Glycolysis makes four ATP during its payoff phase, but it first consumes two ATP. The cell therefore gains two ATP overall. We will return to that accounting after we examine the pathway.

Slide 10 Evolutionary Perspective

Glycolysis is considered an ancient metabolic pathway. The earliest cells were prokaryotic and lacked membrane-bound organelles such as mitochondria. Earth's early atmosphere also contained very little free oxygen. A pathway that required a mitochondrion or abundant oxygen would not have been useful to those early cells.

Glycolysis evolved under those conditions because it occurs in the cytosol and does not require oxygen. Its nearly universal presence across bacteria, archaea, plants, fungi, and animals supports the idea that it developed early and was conserved through evolution. The enzymes may vary somewhat among organisms, but the central pathway is remarkably similar.

This also explains why glycolysis remains important even in organisms with mitochondria. Aerobic respiration adds later stages that capture much more energy, but those stages still depend on glycolysis to convert glucose into pyruvate and to begin transferring electrons to NADH.

Slide 11 Steps of Glycolysis

Although glycolysis contains ten enzyme-catalyzed reactions, we can organize them into three major events. First is the energy-investment phase. The cell uses two ATP to add two phosphate groups to glucose. This phosphorylation raises the molecule's energy and makes it less stable. In other words, the cell must invest energy before it can release and capture more energy.

Second, the unstable six-carbon molecule splits into two three-carbon molecules. After rearrangement, each becomes glyceraldehyde-3-phosphate (G3P). From this point forward, every remaining reaction occurs twice for each original glucose because there are two G3P molecules.

Third is the energy-payoff phase. Each G3P is oxidized. Electrons and hydrogen are transferred to NAD+, reducing it to NADH. The pathway also transfers phosphate groups to ADP through substrate-level phosphorylation. The pathway produces four ATP in total, and the two three-carbon end products become pyruvate. Because two ATP were invested and four were produced, the net gain is two ATP.

Slide 12 Glycolysis Video

As you watch, do not try to memorize the name of every intermediate molecule. Instead, follow four major ideas: where the two ATP are invested, when the six-carbon molecule splits, when NAD+ becomes NADH, and how four ATP are generated during the payoff phase.

Also pay attention to carbon accounting. The pathway begins with one six-carbon glucose and ends with two three-carbon pyruvate molecules. Since carbon dioxide is not produced, all six carbons must still be present. That observation helps distinguish glycolysis from pyruvate oxidation and the Krebs cycle, where carbon dioxide is released.

Slide 13 Glycolysis Summary

Let's complete the accounting. Glycolysis converts one glucose into two pyruvates. It produces four ATP but consumes two ATP, giving a net yield of two ATP. It also produces two NADH molecules. Because no oxygen is required and no carbon dioxide is released, glycolysis can continue in either aerobic or anaerobic conditions as long as the cell can regenerate NAD+.

When oxygen is available, pyruvate enters the mitochondrion for further oxidation. The two NADH molecules carry high-energy electrons toward the electron transport chain. Depending on the cell and the transport system used, shuttle mechanisms transfer electrons from cytosolic NADH into the mitochondrion. At the Honors Biology level, the essential idea is that NADH stores and transports electron energy that can later contribute to ATP production.

The pyruvate molecules still contain most of the original chemical energy from glucose. Glycolysis has begun oxidation, but it has not fully oxidized glucose. This is why aerobic respiration can produce far more ATP than glycolysis alone.

Slide 14 Branching Point

After glycolysis, the pathway reaches a metabolic branching point. If oxygen is available, pyruvate enters the mitochondrion and aerobic respiration continues through pyruvate oxidation, the Krebs cycle, and the electron transport chain. Oxygen is used directly only at the end of the ETC, but its presence keeps the entire aerobic pathway operating because it accepts the electrons.

If oxygen is absent or cannot meet the cell's demand, the electron transport chain cannot accept additional electrons from NADH. The cell then faces an immediate problem: NAD+ becomes scarce. Glycolysis requires NAD+ as an electron acceptor, so glycolysis would stop even if glucose were still available.

Fermentation solves this problem by transferring electrons from NADH to an organic molecule derived from pyruvate. This oxidizes NADH back to NAD+, allowing glycolysis to continue. Fermentation does not produce additional ATP. The two net ATP associated with fermentation come from glycolysis. Its purpose is NAD+ regeneration, not efficient ATP production.

Slide 15 Lactic Acid Fermentation

During lactic acid fermentation, pyruvate accepts electrons from NADH and is reduced to lactate. At the same time, NADH is oxidized to NAD+. The regenerated NAD+ returns to glycolysis, allowing the cell to continue making a net of two ATP per glucose even when oxygen availability is limited. The fermentation reaction itself does not add more ATP.

Animals and some microorganisms can perform this pathway. In human muscle, lactate production increases when ATP demand exceeds the rate at which aerobic respiration can supply it. Lactate is not simply a useless waste product. It can travel to other tissues, where it can be oxidized back to pyruvate, and the liver can use lactate in the Cori cycle to help regenerate glucose.

People often blame the burning sensation during intense exercise entirely on lactic acid, but that explanation is incomplete. The sensation is associated more broadly with changes in muscle chemistry, including hydrogen-ion accumulation and metabolic stress. Lactate production actually helps maintain glycolysis by regenerating NAD+.

Slide 16 Alcohol Fermentation

Alcohol fermentation occurs in yeast and some bacteria. Pyruvate is first decarboxylated, removing one carbon and releasing it as carbon dioxide. The remaining two-carbon compound then accepts electrons from NADH and is reduced to ethanol. As in lactic acid fermentation, NADH is oxidized to NAD+, which returns to glycolysis.

Humans use this pathway commercially. In bread making, carbon dioxide becomes trapped in the dough and causes it to rise; much of the ethanol evaporates during baking. In alcoholic beverage production, ethanol is the desired product, while carbon dioxide may be retained or released depending on the process.

Alcohol fermentation still yields only the two net ATP made during glycolysis. It is also self-limiting because ethanol is toxic to yeast at sufficiently high concentrations. In the simplified pathway shown here, ethanol is an end product and is not converted back to pyruvate to continue respiration.

Closing Statement

Glycolysis provides a small but rapid supply of ATP and produces pyruvate and NADH. When oxygen is available, those products feed aerobic respiration and support much greater ATP production. When oxygen is unavailable, fermentation regenerates NAD+ so glycolysis can continue. The key distinction is that fermentation sustains ATP production indirectly; it does not produce additional ATP beyond glycolysis.